

Lenguaje ensamblador mips 32

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Instruccion	Ejemplo	Significado	Funcionamiento
add	add \$s1,\$s2,\$s3	$\$s1 = \$s2 + \$s3$	Suma
sub	sub \$s1,\$s2,\$s3	$\$s1 = \$s2 - \$s3$	Resta
add immediate	addi \$s1,\$s2,100	$\$s1 = \$s2 + 100$	Suma constante
load word	lw \$s1,100(\$s2)	$\$s1 = \text{Memory}[\$s2 + 100]$	Palabra de memoria a registro
store word	sw \$s1,100(\$s2)	$\text{Memory}[\$s2 + 100] = \$s1$	Palabra de registro a memoria
load half	lh \$s1,100(\$s2)	$\$s1 = \text{Memory}[\$s2 + 100]$	Media palabra de memoria a registro
store half	sh \$s1,100(\$s2)	$\text{Memory}[\$s2 + 100] = \$s1$	Media de registro a memoria
load byte	lb \$s1,100(\$s2)	$\$s1 = \text{Memory}[\$s2 + 100]$	Byte de memoria a registro
store byte	sb \$s1,100(\$s2)	$\text{Memory}[\$s2 + 100] = \$s1$	Byte de registro a memoria
And	and \$s1,\$s2,\$s3	$\$s1 = \$s2 \& \$s3$	And
Or	or \$s1,\$s2,\$s3	$\$s1 = \$s2 \$s3$	Or
Nor	nor \$s1,\$s2,\$s3	$\$s1 = \sim (\$s2 \$s3)$	Nor
shift left logical	sll \$s1,\$s2,10	$\$s1 = \$s2 \ll 10$	Desplaza a la izquierda
shift right logical	slr \$s1,\$s2,10	$\$s1 = \$s2 \gg 10$	Desplaza a la derecha
branch of equal	beq \$s1,\$s2,L	if ($\$s1 == \$s2$) go to L	Igual
branch of not equal	bne \$s1,\$s2,L	if ($\$s1 \neq \$s2$) go to L	Distinto
Branch if Greater Than	bgt \$s1,\$s2,L	if ($\$s1 > \$s2$) go to L	Mayor que
Branch if Greater or Equal	bge \$s1, \$s2, L	if ($\$s1 \geq \$s2$) go to L	Mayor o igual
Branch if Greater Than Zero	bgtz \$s1, L	if ($\$s1 > 0$) go to L	Mayor que cero
Branch if Greater or Equal to Zero	bgez \$s1, L	if ($\$s1 \geq 0$) go to L	Mayor o igual a cero
Branch if Less Than	blt \$s1,\$s2,L	if ($\$s1 < \$s2$) go to L	Menor que
Branch if Less or Equal	ble \$s1, \$s2, L	if ($\$s1 \leq \$s2$) go to L	Menor o igual
Branch if Less Than Zero	bltz \$s1, L	if ($\$s1 < 0$) go to L	Menor que cero
Branch if Less or Equal to Zero	blez \$s1, L	if ($\$s1 \leq 0$) go to L	Menor o igual a cero
Jump	j 2500	go to 10000	Salto a la direccion